

Semester/Year End Examination May-2023

Name of School: **School of Engineering and Technology**

Name of Programme/Course: **B.Tech. (Biomedical) / B.Tech. (Biotech) / B.Tech. (Chemical) / B.Tech. (Civil) / B.Tech. (Mechanical) / B.Tech. (CSE) / B.Tech. (CSE-AIML) / B.Tech. (CSE-Cyber Security) / B.Tech. (Food Tech.) / B.Tech. (Electrical)**

Semester: **II**

Paper Code: **BTESSDS202T22**

Title of Paper: **Smart and Sustainable Energy**

Time: **3 hrs.**

Max. Marks: **60**

INSTRUCTIONS/NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section-A and Section-B. Section-A consists of seven questions and Section-B consists of six questions. Attempt any 5 questions from Section-A and any 4 questions from Section-B. Marks allotted are mentioned against each question.

Question Number	SECTION-A	Marks Allotted
1.	Write some important uses of solar energy. - ④	4
2.	Write a short note on energy consumption in India.	4
3.	How does biomass energy work? - ②	4
4.	Write a short note on distributed generation? - ②	4
5.	What are fossil fuels? How fossil fuel is used as alternative energy source? ③	4
6.	What is geothermal energy? Write its advantages and disadvantages. - ④	4
7.	Discuss different types of Tidal energy generators.	4

SECTION-B

1. What is energy? Explain its classification. — (10) 10
2. What is hydroelectric energy? Discuss its advantages and disadvantages. — (5) 10
3. What is energy efficiency? Discuss green building. — (10) 10
4. What are nonrenewable source of energy? Explain its types. 10
5. What do you understand by energy transportation and distribution? 10
6. What is wind energy? Explain in detail. (5) (5) 10